

$$P = \sqrt{33}$$

Ficha 2

1. a)

$$\begin{bmatrix} 0 & 1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ -1 & -1 & 0 & 1 \\ -1 & -1 & -1 & 0 \end{bmatrix}$$

b)

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

c)

$$\begin{bmatrix} 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \\ 1 & -1 & 1 & -1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

2. a) $BA =$

$$\begin{bmatrix} -1 & 3 \\ 2 & 10 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 3 & 10 \\ 2 & 0 & -2 \\ 5 & 2 & 5 \end{bmatrix}$$

$$b) AB - 3C = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 3 \\ 2 & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 0 & -3 \end{bmatrix}$$

$$= \begin{bmatrix} -2 & 1 \\ 3 & 12 \end{bmatrix} - \begin{bmatrix} 3 & 3 \\ 0 & -3 \end{bmatrix} = \begin{bmatrix} -5 & -2 \\ 3 & 15 \end{bmatrix}$$

$$c) \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$d) \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & -1 \end{bmatrix} = \begin{bmatrix} 2 & 0 & -1 \\ 4 & 0 & -2 \\ 6 & 0 & -3 \end{bmatrix}$$

$$e) \begin{bmatrix} 2 & 0 & -1 \end{bmatrix} \times \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -1 \end{bmatrix}$$

$$f) \begin{bmatrix} -1 & 3 \\ 2 & 0 \\ 1 & 2 \end{bmatrix} \times \begin{bmatrix} -1 & 2 & 1 \\ 3 & 0 & 2 \end{bmatrix} = \begin{bmatrix} 10 & -2 & 5 \\ -2 & 4 & 2 \\ 5 & 2 & 5 \end{bmatrix}$$

$$3. a) A = I_m = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1+x & x & x \\ x & 1+x & x \\ x & x & 1+x \end{bmatrix} = \begin{bmatrix} 2x-2 & 2x-2 & 2x-2 \\ 2x-2 & 2x-2 & 2x-2 \\ 2x-2 & 2x-2 & 2x-2 \end{bmatrix} \quad (*)$$

$$X = \begin{bmatrix} -1/3 & -1/2 & -1/2 \\ -1/2 & -1/3 & -1/2 \\ -1/2 & -1/2 & -1/3 \end{bmatrix}$$

$$b) \begin{bmatrix} 3x + \frac{3}{2} & 3x & 3x \\ 3x & 3x + \frac{3}{2} & 3x \\ 3x & 3x & 3x + \frac{3}{2} \end{bmatrix} = \begin{bmatrix} 5x + \frac{15}{4} & 5x + \frac{15}{4} & 5x + \frac{15}{4} \\ 5x + \frac{15}{4} & 5x + \frac{15}{4} & 5x + \frac{15}{4} \\ 5x + \frac{15}{4} & 5x + \frac{15}{4} & 5x + \frac{15}{4} \end{bmatrix}$$

$$X = \begin{bmatrix} -\frac{9}{8} & -\frac{15}{8} & -\frac{15}{8} \\ -\frac{15}{8} & -\frac{9}{8} & -\frac{15}{8} \\ -\frac{15}{8} & -\frac{15}{8} & -\frac{9}{8} \end{bmatrix}$$

$$4. a) AB = \begin{bmatrix} 1 & 4 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} -2 & 4 & 0 \\ 2 & -1 & 0 \\ 2 & -1 & -3 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & +6 \end{bmatrix}$$

$$b) AX = Im$$

$$\Rightarrow \begin{bmatrix} 1 & 4 & 0 \\ 2 & 2 & 0 \\ 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_{11} & x_{12} & x_{13} \\ x_{21} & x_{22} & x_{23} \\ x_{31} & x_{32} & x_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{e)}$$

$$\Rightarrow \begin{bmatrix} x_{11} + 4x_{21} & x_{12} + 4x_{22} & x_{13} + 4x_{23} \\ 2x_{11} + 2x_{21} & 2x_{12} + 2x_{22} & 2x_{13} + 2x_{23} \\ 2x_{21} - 2x_{31} & 2x_{22} - 2x_{32} & 2x_{23} - 2x_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\left\{ \begin{array}{l} x_{11} = x_{31} \\ x_{12} = x_{32} \\ x_{11} = -x_{21} \\ x_{13} = -x_{23} \\ \dots \\ \dots \\ \dots \\ \dots \end{array} \right. \quad \text{Impossible!}$$

$$5. a) \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} //$$

$$B) \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{matrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{21} & \alpha_{22} & \alpha_{23} \\ \alpha_{31} & \alpha_{32} & \alpha_{33} \end{matrix} = \begin{bmatrix} 1 & 0 & 2 \\ 1 & -1 & 1 \\ -2 & 1 & 0 \end{bmatrix} \Leftrightarrow$$

$$\Leftrightarrow \begin{cases} \alpha_{11} = 1 \\ \alpha_{12} = 0 \\ \alpha_{13} = 2 \\ \alpha_{21} = 0 \\ \alpha_{22} = -1 \\ \alpha_{23} = -1 \\ \alpha_{31} = -3 \\ \alpha_{32} = 0 \\ \alpha_{33} = -1 \end{cases}$$

$$6. a) AB = \begin{bmatrix} 2 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$b) AB = 0 \Leftrightarrow$$

$$\Leftrightarrow A^{-1} A B = 0 A^{-1} \Leftrightarrow$$

$$\Leftrightarrow \text{Im } B = 0 \Leftrightarrow$$

$$\Leftrightarrow B = 0,,$$

$$c) \left[\begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 2 & -1 & 0 & 1 \end{array} \right] = \left[\begin{array}{cc|cc} 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right]$$

não é invertível

$$7. a) A = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix}$$

$$B = \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix}$$

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

$$AC = \begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 5 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$$

$$6) AB = AC \Leftrightarrow$$

$$\Leftrightarrow A^{-1}AB = A^{-1}AC \Leftrightarrow$$

$$\Leftrightarrow \ln B = \ln C \Leftrightarrow$$

$$\Leftrightarrow B = C$$

$$c) \begin{vmatrix} 0 & 1 \\ 0 & 2 \end{vmatrix} = 0$$

Per isso m é invertível

$$8. a) \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 2 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & -2 & 0 & 1 \end{array} \right]$$

$$b) \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 1 & 3 & 1 & 0 & 0 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & +2 & 1 & 0 \\ 0 & 3 & 1 & -1 & 0 & 1 \end{array} \right] \rightarrow$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -3 & 1 \end{array} \right]$$

$$9. (A+B)^2 = A^2 + 2AB + B^2 \Leftrightarrow$$

$$\Leftrightarrow A^2 + AB + BA + B^2 = A^2 + 2AB + B^2$$

$$\text{Se } AB = BA :$$

$$2AB = 2AB, \quad \text{c.q.m.}$$

$$10. a) (A^T)^T = A$$

$$b) (AC)^T = C^T A^T$$

$$c) (B^T C + D^T C)^T = C^T (B + D)$$

11. Se $A = A^T$ então A é uma matriz simétrica, e se $A \times A^T = A^2$,
 $A \times A^T$ é uma matriz simétrica.

$$12. A^T = A^{-1}$$

$$B^T = B^{-1}$$

$$A \times A^T = \ln$$

$$B \times B^T = \ln$$

$$(A \times B)(A \times B)^T = \ln \Leftrightarrow \ln = \ln$$